

Exercise 01 (7 points)

Let f be the map defined as follows :

$$\begin{aligned} f : \mathbb{N} - \{1\} &\longrightarrow \mathbb{R} \\ n &\longmapsto \frac{n^2 - n + 2}{n - 1}. \end{aligned}$$

1. Compute $f(A)$, $f^{-1}(B)$, $f(C)$ and $f^{-1}(\mathbb{Z})$ where

$$A = \left\{ \frac{1}{3}, 4 \right\}, \quad B =]-\infty, -1] \quad \text{and} \quad C = \{n \in \mathbb{N} - \{1\} : |n^3 - n + 1| \leq 1\}.$$

2. Let H be the following set

$$H = \left\{ x \in \mathbb{Q} : \frac{1}{x - E(x)} \in \mathbb{N} \right\}.$$

Show that $f^{-1}(H) = \{2n + 1 : n \in \mathbb{N}, n \geq 2\}$.

Exercise 02 (6 points)

1. Let R be the set defined as follows :

$$R = \left\{ \frac{a}{10^n} \mid a \in \mathbb{Z}, n \in \mathbb{N} \right\}.$$

- Show that R is a subring of $\mathbb{Q}$ for the usual binary operations.
 - Show that 3 has no inverse in R . Conclude.
 - Determine the set of invertible elements in R .
2. Determine the smallest subring of $\mathbb{Q}$ that contains $\frac{1}{2}$.
3. (a) Find all ring homomorphisms from $\mathbb{Z}$ to $\mathbb{Z}/30\mathbb{Z}$.
- (b) In each case describe the kernel and the image.

Exercise 03 (7 points)

Let $P(X) = \sum_{i=0}^n a_i X^i \in \mathbb{Z}[X]$, where $a_0 \neq 0$ and $a_n \neq 0$.

- Suppose that P has a root $r \in \mathbb{Q}$, such that $r = \frac{p}{q}$ with $(p, q) \in \mathbb{Z}^* \times \mathbb{N}^*$. Show that $p|a_0$ and $q|a_n$.
- Give the factorization in $\mathbb{Z}[X]$ of $Q(X) = 2X^3 - X^2 - 13X + 5$.
- Let $R(X) = X^3 + 3X - 1$ and $S(X) = X^4 + X^2 + 1$. Are $R(X)$ and $S(X)$ irreducible polynomials in $\mathbb{Q}[X]$?
- Split the following fractions into sums of partial fractions in $\mathbb{C}(X)$ and in $\mathbb{R}(X)$:

(a) $\frac{1}{X^3 - X}.$

(b) $\frac{X^2 + 2X + 5}{X^2 - 3X + 2}.$

Good luck